

NOVA: A Self-Supervised Graph Framework for Real-Time Anomaly Detection in Internet of Vehicles

Arash Heidari, Jamal N. Al-Karaki

Abstract— The Internet of Vehicles (IoV) enables cooperative driving and real-time Vehicle-to-Everything (V2X) communication but remains vulnerable to behavioral and structural anomalies due to its dynamic, decentralized nature. Existing deep learning methods either overlook topological inconsistencies or ignore communication feature fidelity, while random-walk sampling introduces contextual noise. In this paper, we propose Network Observation for Vehicular Anomalies (NOVA), a self-supervised graph-based framework that detects both behavioral and structural anomalies in IoV networks without labeled data. NOVA models vehicular communications as attributed graphs and employs intimacy-guided subgraph sampling to extract meaningful neighborhoods. A Graph Convolutional Network (GCN)-based generative module reconstructs node attributes to reveal behavioral deviations, while a contrastive module validates structural coherence through embedding comparisons of real and perturbed contexts. Their hybrid anomaly score enables accurate, scalable, and real-time detection of compromised nodes. Performance results show that NOVA achieves state-of-the-art performance (98.7% accuracy, 98.1% F1), real-time throughput (~4.7k events/s at 5k msg/s), and strong robustness (AUROC 0.99, AUPRC 0.98, FAR 0.05) with near-linear scalability (≤ 40 ms latency for 50k vehicles). By integrating generative and contrastive self-supervised learning with context-aware sampling, NOVA significantly enhances IoV security, reliability, and adaptability.

Index Terms— Internet of Vehicles, V2X Security, Anomaly Detection, Self-Supervised Learning, Graph Neural Networks.

I. INTRODUCTION

The Internet of Vehicles (IoV) has become a key part of Intelligent Transportation Systems (ITS) [1, 2]. It lets vehicles, Roadside Units (RSUs), and cloud infrastructures connect easily through Vehicle-to-Everything (V2X) communication [3]. This high level of connectivity makes it possible for safety-critical applications like cooperative adaptive cruise control, autonomous driving coordination, and sending emergency messages [4]. IoV

communication is dynamic and decentralized, which makes it easy for anomalous behaviors and cyber-physical threats to happen [5, 6]. Nodes that do not work right or are hacked can send bad data, stop routing updates, or change transmissions that are sensitive to delay, which can have serious effects on road safety and traffic flow [7]. So, it is very important to build strong frameworks for detecting anomalies to keep IoV ecosystems safe, reliable, and strong [8, 9].

Recent developments in deep learning have provided powerful tools for anomaly detection in the IoV, as their capability to learn intricate spatiotemporal correlations in vehicular data has been effectively leveraged [10, 11]. On the other hand, Graph Neural Networks (GNNs), Convolutional Neural Networks (CNNs), and hybrid deep architectures have been utilized to simulate the communication dynamics inside IoV environments [12]. These models use things like packet transmission rates, mobility patterns, and changes in topology to find unusual behaviors [13, 14]. However, current methods frequently encounter two significant challenges: (1) the difficulty in detecting both node-level behavioral anomalies and structural irregularities in dynamic vehicular graphs, and (2) an excessive dependence on labeled training datasets, which are expensive and impractical in extensive vehicular contexts [15, 16]. Consequently, there is an urgent requirement for self-supervised and hybrid anomaly detection methodologies capable of generalizing across diverse IoV scenarios without extensive labeling [6, 17, 18].

Although significant progress has been achieved, there are still significant gaps that hinder trust in real-world IoV anomaly detection. In dynamic V2X graphs, most techniques identify behavioral or structural irregularities, but not both. Under the mobility churn, the common neighborhood sampling methods can contain noise, leading to unstable representations and false alarms. Furthermore, there are many studies that focus on the accuracy of the data but do not provide any real-time guarantees of latency and throughput in large, safety-critical V2X deployments. Transferable adversarial attacks are also an emerging threat to deep IDS models, as perturbations generated on surrogate models may transfer to target detectors by exploiting shared latent feature representations [19]. Network Observation for Vehicular Anomalies (NOVA) is made to fill these gaps by combining generative attribute reconstruction with node-context contrastive structural validation and using intimacy-guided subgraph sampling to keep high-fidelity neighborhoods while keeping a scalable pipeline that can be used in real time. NOVA sees vehicular networks as topological graphs, with vehicles, RSUs, or infrastructure as nodes and Vehicle-to-Vehicle (V2V) / Vehicle-to-Infrastructure (V2I) / Vehicle-to-Network (V2N) communication as edges. NOVA learns to

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find both behavioral and topological anomalies without any help by using subgraph sampling, attribute reconstruction, and structural contrastive learning. It combines attribute-level reconstruction errors and structure-level consistency scores into a hybrid anomaly score during inference. This makes it possible to find compromised or malfunctioning nodes in real time and on a large scale. NOVA combines generative reconstruction and contrastive validation, but its novelty does not come from having two self-supervised losses at the same time. Instead, it comes from a non-trivial combination of (i) intimacy-guided subgraph sampling that filters out low-fidelity neighbors caused by mobility and stabilizes the node's contextual view, (ii) target-node anonymization within sampled contexts, which stops shortcut learning and forces true context-conditioned inference (which makes it more robust to poisoning and transient correlations), and (iii) a hybrid anomaly scoring mechanism that explicitly combines attribute deviation signals with topology-consistency signals. These parts are all connected in important ways. For example, without intimacy-guided sampling, contrastive node-context validation is dominated by noisy contexts. Without anonymization, reconstruction can turn into feature copying. And without hybrid scoring, either branch alone is systematically blind to a major anomaly class. This synergy lets NOVA find small problems that are both behavioral and structural, and it stays stable even when there is a lot of change and can grow to handle real-time IoV operations. This paper's primary contributions can be summarized as follows:

- Modeling IoV communication as a dynamic attributed graph that includes both operational aspects and topological connection;
- Introducing a new hybrid framework that uses generative learning to rebuild attributes and contrastive learning to make sure that structures are consistent;
- Creating a better subgraph sampling technique based on closeness scoring to get high-quality communication situations;
- Making a dual-path self-supervised learning pipeline that does not need labeled training data;
- Showing that NOVA works by doing both theoretical analysis and real-world testing, which shows that it can better find both behavioral and structural problems in IoV networks.

The rest of the paper is organized as follows. Section II presents a summary of the latest findings that are related to the study topic. The system model is elaborated in Section III. Also, the NOVA method will be provided in Section IV. The paper presents the results of the NOVA method and provides a comprehensive comparison with existing approaches in Section V. Section VI provides an overview and conclusion.

II. RELATED WORK

In this section, we consider related research addressing the topic of finding anomalies and intrusions in the IoV and linked vehicle systems. In this regard, Sboui, et al. [20] presented a

Convolutional Neural Network–Bidirectional Long Short-Term Memory (CNN-BiLSTM) / Variational Autoencoder (VAE) framework for detecting anomalies in spatio-temporal Light Detection and Ranging (LiDAR) data. As well, Tiwari, et al. [21] also suggested a fine tree-based IDS for IoV, which achieved high accuracy on benchmark datasets. In addition, Cheng, et al. [22] used a spatial-temporal correlation module with attention and a patch-based transformer. In a similar project, Cao, et al. [23] created a self-supervised IDS for In-Vehicle Networks (IVNs) that used hierarchical transformers and can be updated online. As well, A hybrid model was proposed by Zhao, et al. [24] to find problems in Basic Safety Messages (BSMs). It combined VAE and Generative Adversarial Network (GAN). Singh and Rathore [25], on the other hand, used a Bayesian Online Change Point identification (BOCPD) method for real-time anomaly identification in connected vehicles, along with ways to recover from them. Moreover, Li, et al. [26] introduced a meta-heuristic optimized deep random neural network that was 99.2% accurate at finding IoT-based anomalies. Mansourian, et al. [27] also suggested a Controller Area Network (CAN) bus IDS that used LSTM and ConvLSTM modules along with a Gaussian Naïve Bayes classifier. Plus, Van Wyk, et al. [28] used a χ^2 -detector with CNN and Kalman filtering to find anomalies in Connected and Autonomous Vehicles (CAVs). Devika, et al. [29] made a GAN-based anomaly detector that works best for CAV dynamics. As well, Yang, et al. [30] put forward an approach for detecting GPS spoofing attacks that is based on domain knowledge and uses trajectory prediction and statistical dissimilarity tests. Moreover, He, et al. [31] introduced a multi-information fusion CNN that focuses on vehicle sensor abnormalities. In another work, He, et al. [32] created a Wavelet Kernel Network with omni-scale convolution that picked up high-frequency information for finding anomalies in ITS. Khanmohammadi and Azmi [33] also came out with a D-CNN-LSTM Autoencoder that used preprocessing approaches like Differencing and Moving Standard Deviation to greatly increase F1-scores. Also, Zhang, et al. [34] suggested a dual-channel self-attention CNN for finding anomalous patterns in multivariate time series data. Al-Absi, et al. [35] presented the DST-IDS, a dynamic spatial-temporal graph-transformer network to detect intrusions in IVNs, which captured the changing nature of vehicle communication patterns. Lastly, Wang, et al. [36] proposed a class-imbalance-aware IDS on spatiotemporal GNNs on software-defined vehicles that learned the topology and temporal communication structures.

There are a lot of different ways to find anomalies in IoV and CAVs, but most of them have serious flaws. In real-world driving situations, labeled datasets are hard to find and expensive; thus, many supervised methods need them. Generative and hybrid models make it easier to find anomalies, but they often have problems with scalability and training stability. Self-supervised and transformer-based algorithms lessen the need for labeling, but they mostly concentrate on modeling sequential messages, ignoring the structural abnormalities that are typical in IoV networks. Also, sampling methods like random walks often introduce noise

and make detection less accurate. To address these issues, we provide NOVA, a self-supervised graph-based framework that incorporates intimacy-guided subgraph sampling, generative Graph Convolutional Network (GCN) reconstruction, and contrastive structural validation. NOVA strikes a balance between communication fidelity and structural consistency, creating a single anomaly score that may be used to find compromised nodes in real time. This makes it scalable, label-free, and able to handle both behavioral and structural problems, which are the main problems with earlier studies.

III. SYSTEM MODEL

In NOVA, the IoV communication environment is shown as a topology graph. Each node represents an entity in the vehicular network, like a vehicle, RSU, or infrastructure component. Each edge shows an active communication link, such as V2V, V2I, or V2N connections. The communication and operational features of each node, like packet transmission rates, latency, signal strength, routing updates, and mobility patterns, make up its attribute vector. On-board sensors and network monitoring modules constantly gather these parameters and send them to a security management system for centralized or edge-based analysis. The structural half of the graph shows the changing communication topology of the vehicular network, while the attribute part shows how each participant behaves and works. This changes the problem of finding malicious or malfunctioning activity into the problem of finding nodes in the graph that are out of the ordinary. NOVA uses an augmented intelligence-driven detection technique to find these kinds of problems. The network is formally represented as $G = (V, E, X, A)$, where $V = \{v^1, v^2, \dots, v_n\}$ constitutes the set of nodes, E represents the set of edges, $X \in \mathbb{R}^{N \times D}$ signifies the node feature matrix, and $A \in \mathbb{R}^{N \times N}$ denotes the adjacency matrix, with D indicating the number of feature dimensions. Fig. 1 shows how the NOVA engine works in an IoV setting, where vehicles share information over V2V links and talk to RSUs that send data to a security management system to find problems. The green dots show safe vehicles, the red dots show nodes that are not normal, and the blue dashed lines depict V2V communications. Also, Table I shows the used symbols.

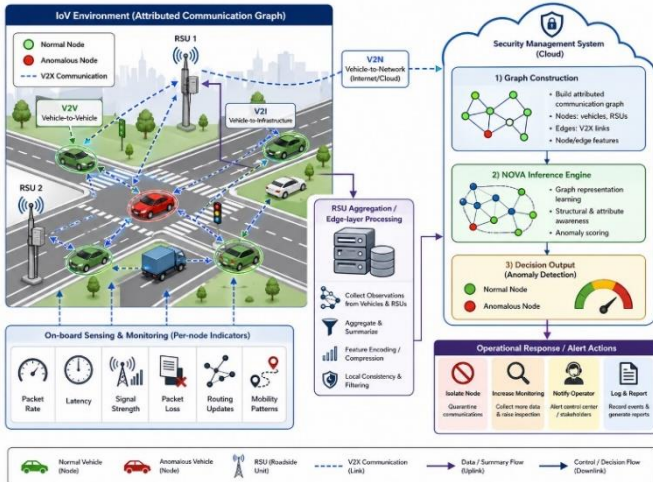


Fig. 1. NOVA system model for IoV anomaly detection.

TABLE I
DEFINITIONS OF THE SYMBOLS

Symbol	Definition	Symbol	Definition
G	Communication graph	V	Set of nodes
E	Set of edges	X	Node feature matrix
A	Adjacency matrix of the vehicular graph	D	Number of feature dimensions
$g_i^{\phi_j}$	j -th contextual subgraph	S	Communication relevance matrix
α	Damping factor	I_i^t	Intimacy score
$\bar{A}$	Column-normalized adjacency matrix	H^{ϕ_j}	Node embedding matrix of the subgraph
$L_c^{\phi_j}$	Contrastive loss for	x_i	Original feature vector
$\hat{x}$	Reconstructed feature vector	$Enc(\cdot), Dec(\cdot)$	Encoder and decoder functions
$\sigma(\cdot)$	Nonlinear activation function	$\hat{x}_i^{\phi_j}$	Reconstructed feature matrix
h_i	Embedding vector of target node v_i	s_j	Pooled embedding vector
e_i^j	Positive pair:	$\bar{e}_i^j$	Negative pair
λ, γ	Weighting factors	θ	Scaling parameter
$score(v_i)$	Final anomaly score	L	Overall training loss

IV. THE NOVA METHOD

NOVA identifies abnormal IoV objects in three synchronized processes: intimacy-directed context selection, GCN-based attribute reconstruction and node-context contrastive validation. Given each target node, NOVA first identifies high-relevance subgraphs of context, masks the target-node features within each context, and then measures whether the node fits in the communication neighborhood of its attributes and structures. The overall methodological pipeline is summarized in Fig. 2.

Next, NOVA uses graph-view sampling to get subgraphs that show the local communication neighbourhood of each target node. This is important because high-quality context is needed to learn reliable representations, and naive random-walk sampling can add noise by treating all neighbours as equally informative. Specifically, NOVA first uses random walks to create several candidate subgraphs. Then, it uses a custom PageRank-inspired scoring system to figure out how important each neighbour is by calculating the relevance matrix $S = \alpha I - (1 - \alpha)\bar{A}$, where I is the identity matrix, $\alpha \in [0,1]$ (typically 0.15), and $\bar{A} = AD^{-1}$ is the column-normalized adjacency matrix with $D(i,i) = \sum_j A(i,j)$. Here, $S(i,j)$ measures how well v_i and v_j communicate with each other. Each candidate subgraph $g_i^{\phi_j}$ gets an intimacy score $I_i^t = \sum_j S(i,j)$. The top- k subgraphs with the highest intimacy scores are kept as the best training contexts. Finally, NOVA uses target-node anonymization by zero-masking the original features of the target node in each sampled subgraph. This stops direct information leakage and makes the model guess node properties and structural changes based on the communication context around them. This makes it easier to find both behavioural and topological anomalies.

Formally, for a selected contextual subgraph $g_i^{\phi_j}$ of target node v_i with feature matrix $X_i^{\phi_j} \in \mathbb{R}^{|V_s| \times F}$, NOVA performs

target-node anonymization by replacing the target node's feature vector with an all-zero vector, $\tilde{x}_i = 0 \in \mathbb{R}^F$. The resulting masked feature matrix $\tilde{X}_i^{(\phi_j)}$ is defined by setting the target row to zero while leaving all other node features unchanged. So, NOVA uses explicit zero-masking rather than a dataset-mean vector or a learnable [MASK] token.

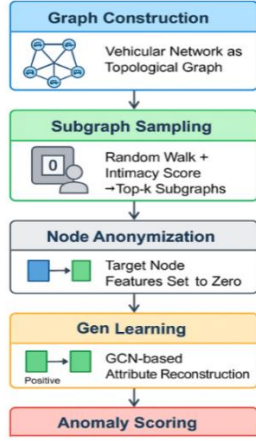


Fig. 2. NOVA pipeline.

Random-walk and k-hop sampling tend to include too many transient or weakly related neighbors in dynamic IoV graphs. This makes the context noise higher: when links appear and disappear quickly, the sampled neighborhood distribution changes between time windows, making subgraphs with higher variance and less consistent supervision for both reconstruction and contrastive alignment. Intimacy-guided sampling helps with this by ranking possible contexts based on a communication-relevance signal (PageRank-inspired intimacy). This works like a relevance filter that focuses probability mass on stable, high-interaction neighbors. Churn lowers the expected number of temporary edges and the variance of the aggregated context embedding. This means that the generative branch sees less reconstruction noise (more predictable neighbor-conditioned attributes), and the contrastive branch sees a clearer separation between real and corrupted contexts. In contrast, naive k-hop expansion can make neighborhoods bigger in dense areas and make hub-induced dilution worse, while random walks can go into loosely connected areas and add nodes that are not in the right context. Importantly, intimacy-guided selection makes the system more robust without raising the asymptotic cost. This is because sampling stays local and scoring candidates is limited by the same neighborhood statistics that are already available at RSUs. This creates a useful trade-off: lower-noise contexts and more stable learning when mobility churn happens. This explains the empirical FAR/F1 gains over RandomWalk and degree-based baselines.

NOVA uses a one-layer GCN encoder to map the node features of each subgraph into a low-dimensional embedding space:

$$H^{\phi_j} = \text{Enc}_{\text{GCN}}(X_i^{\phi_j}, A_i^{\phi_j}) \quad (1)$$

$$= \sigma \left(D^{-\frac{1}{2}} (A_i^{\phi_j} + I) D^{-\frac{1}{2}} X_i^{\phi_j} W_e \right)$$

Here, H^{ϕ_j} is the node embedding matrix of the subgraph $g_i^{\phi_1}$, $X_i^{\phi_j}$ and $A_i^{\phi_j}$ are the feature and adjacency matrices of the subgraph, respectively, $\sigma(\cdot)$ denotes a non-linear activation function such as ReLU, $D(i, i) = \sum_j A(i, j)$ is the degree matrix, and W_e is the learnable weight matrix for the encoder.

If fewer than k valid high-intimacy contexts are available for a target node, NOVA uses all available contexts $m_i < k$ for that node and normalizes the generative and contrastive objectives by m_i instead of k . No synthetic padding is used during inference; during batched training, missing slots are masked out and excluded from loss aggregation.

The graph decoder in NOVA is set up the same way as the encoder, using one GCN layer to recreate the original node attributes from the learned embeddings. The decoder uses the embedding matrix that the encoder made as its input. The decoder sends information over the subgraph's local communication topology. The decoding process is defined as:

$$\tilde{X}_i^{\phi_j} = \text{Dec}_{\text{GCN}}(H^{\phi_j}, A_i^{\phi_j}) \quad (2)$$

$$= \sigma \left(D^{-\frac{1}{2}} (A_i^{\phi_j} + I) D^{-\frac{1}{2}} H^{\phi_j} W_d \right)$$

where W_d is the weight matrix that can be trained for the decoding process, H^{ϕ_j} is the matrix that the encoder uses to incorporate, $A_i^{\phi_j}$ is the subgraph's adjacency matrix, D is the degree matrix, and $\sigma(\cdot)$ is a nonlinear activation function like ReLU.

For a certain target node v_i attribute reconstruction depends on its anonymized subgraphs $g_i^{\phi_1}, \dots, g_i^{\phi_k}$. NOVA makes the reconstruction process reliant only on contextual information from adjacent nodes by setting the target node's characteristics to zero. This architecture ensures that both direct neighbors and connections of a higher order help find behavioral inconsistencies. The goal of the generative learning process is to lower the Mean Squared Error (MSE) between the reconstructed feature vector and the target node's original feature vector. The generating loss for the j th selected subgraph is:

$$L_g^{\phi_j} = \frac{1}{N} \sum_{i=1}^N (\tilde{X}_i^{\phi_j}[-1, :], x_i)^2, j \in \{1, \dots, k\} \quad (3)$$

where $\tilde{X}_i^{\phi_j}$ is the reconstructed feature matrix from the decoder for subgraph $g_i^{\phi_j}$, and x_i is the initial feature vector for node v_i . To find the ultimate generating loss for a target node, take the average of all k chosen subgraphs:

$$L_g = \frac{1}{k} \sum_{j=1}^k (L_g^{\phi_j}) \quad (4)$$

Although the generative branch can be used to capture the attribute inconsistency, structural anomalies need to be compared between the target node and its communication context. NOVA thus incorporates the contrastive branch, which matches target embedding to its actual contextual subgraph and distinguishes between it and degraded contextual views.

The purpose of the contrastive learning module in NOVA is to find structural problems by comparing the representation

of a target node with that of its contextual subgraph. The GCN encoder takes in two parts: the feature matrix of the subgraph and the feature vector of the target node. The way to encode the feature matrix of the subgraph is the same as what was said before in Eq. (1). For individual nodes, the encoding method is more straightforward since a single node lacks its adjacency structure. The embedding of a target node v_i is gotten as:

$$h_i = \sigma(x_i W_e) \quad (5)$$

where x_i is the feature vector of v_i , h_i is the embedding vector that comes out of it, W_e is the parameter matrix that the generative encoder uses, and $\sigma(\cdot)$ is a nonlinear activation function like ReLU. Also, the parameter matrix W_e is the same as Eq. (1). NOVA uses a pooling function to combine the embeddings of all the nodes in the sampled subgraph into a single representation vector. There are many different pooling strategies, like average pooling, max pooling, and bilinear pooling, but NOVA uses a simple and effective average pooling operation:

$$s_j = \Phi(H^{\varphi_j}) = \frac{1}{n} \sum_{k=1}^n H^{\varphi_j}[k, :] \quad (6)$$

In this case, n is the number of nodes in the subgraph, $g_i^{\varphi_j}$ is the embedding matrix that the GCN encoder makes, and s_j is the vector that represents the generated subgraph. The positive example e_i^j is made by combining the target node's embedding with the real representation of its contextual subgraph:

$$e_i^j = (h_i, s_j), \quad (7)$$

The negative example $\bar{e}_i^j$ is made by messing up the subgraph embedding, more especially by messing up the matrix that embeds the subgraph H^{φ_j} and then putting it all back together using the pooling operation $\bar{e}_i^j = (h_i, \bar{s}_j)$. Then, NOVA calculates discrimination ratings for each pair:

$$\begin{aligned} d_i^j &= \sigma(h_i W_d s_j) \\ \bar{d}_i^j &= \sigma(h_i W_d \bar{s}_j) \end{aligned} \quad (8)$$

where W_d is a trainable parameter matrix and $\sigma(\cdot)$ is the sigmoid activation function, ensuring the scores lie within $[0,1]$. An anomalous node should exhibit a much larger difference between d_i^j (positive pair score) and $\bar{d}_i^j$ (negative pair score). The Jensen–Shannon divergence is used by NOVA to quantify this difference:

$$L_c^{\varphi_j} = -\frac{1}{2N} \sum_{i=1}^N \left(\log(d_i^{\varphi_j}) + \log(1 - \bar{d}_i^{\varphi_j}) \right), j \in \{1, \dots, k\} \quad (9)$$

In this case, $L_c^{\varphi_j}$ is the subgraph's contrastive loss for $g_i^{\varphi_j}$. Then, By averaging across all k subgraphs, the node's final contrastive loss is determined:

$$L_c = \frac{1}{k} \sum_{j=1}^k (L_c^{\varphi_j}) \quad (10)$$

The formulation of the generative loss and contrastive loss employed in the training phase of NOVA was covered in the preceding sections. These elements are utilized to calculate the final anomaly scores for every node in the IoV communication graph during the inference stage. Given that most nodes in the network should act appropriately, the trained GCN encoder–decoder should be able to accurately reconstruct and map their feature vectors into a suitable latent representation. On the other hand, when projected into the latent space, anomalous nodes, whether they have structural or attribute irregularities, will have distorted representations, leading to weaker alignment with their context or higher reconstruction errors. Reconstructing a target node's attributes in the generative module only uses the contextual neighbors' features in the chosen subgraphs. An indicator of anomaly likelihood is the degree of discrepancy between the original and reconstructed feature vectors. This discrepancy is measured by NOVA using the L2-average norm distance across the top k contextual subgraphs:

$$\text{score}_g(v_i) = \frac{1}{k} \sum_{j=1}^k (\theta(\|\bar{X}_i^{\varphi_j}[-1, :] - x_i\|_2^2)) \quad (11)$$

where $\text{score}_g(v_i)$ is the generative anomaly score for node v_i , θ is a scaling parameter to normalize scores to $[0,1]$, $\bar{X}_i^{\varphi_j}$ is the reconstructed feature matrix, and x_i is the original feature vector. Higher values of $\text{score}_g(v_i)$ indicate greater deviations in communication behavior relative to its neighborhood. For the contrastive module, anomaly scoring is based on the discrimination scores computed during training for positive and negative sample pairs, as defined before. The contrastive anomaly score is calculated as $\text{score}_c(v_i) = \frac{1}{k} \sum_{j=1}^k (d_i^j - \bar{d}_i^j)$. Here, d_i^j matches the similarity between the target node and its real contextual subgraph, while $\bar{d}_i^j$ is equal to how similar the node is to a broken subgraph. If a node is different from the rest of its neighborhood, it will make d_i^j and $\bar{d}_i^j$ near in number, which makes their difference less. For a regular node, on the other hand, d_i^j will get close to 1 while $\bar{d}_i^j$ gets closer to 0, which makes the difference closer to 1.

Finally, NOVA combines the attribute-based and structure-based signals into one hybrid anomaly score $\text{score}(v_i) = \lambda \text{score}_g(v_i) + \gamma \text{score}_c(v_i)$. The values of λ and γ control how much behavioural deviations and topological inconsistencies contribute to the score. It also optimizes a single objective $L = \lambda L_g + \gamma L_c$, where L is the overall loss that balances generative attribute reconstruction and contrastive structural validation. During training, NOVA first picks a group of target nodes from the IoV graph. For each target node, it makes k candidate contextual subgraphs and keeps the ones with the highest intimacy scores for later dual-

branch learning. Also, Fig. 3 shows NOVA's self-supervised learning approach in detail.

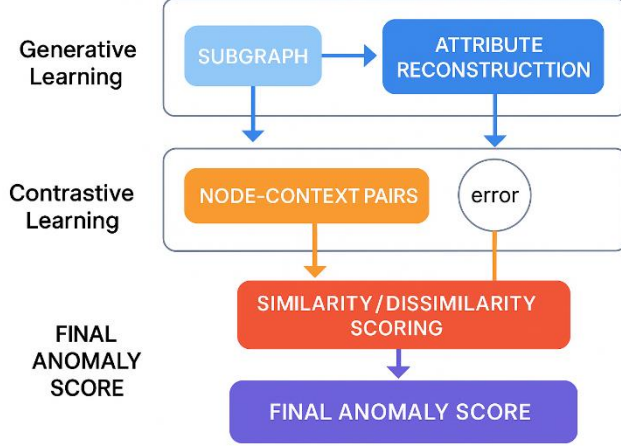


Fig. 3. Dual-path self-supervised learning in NOVA.

V. RESULTS AND COMPARISONS

In the next several parts, we will talk about the simulation environments, datasets, assessment measures, and how our results compare to the best ones out there.

A. Simulation Environments

We create a multi-layer SUMO-NS3 co-simulation platform to test NOVA in practice IoV scenarios with V2V, V2I and V2N communications. SUMO simulates urban, highway, and mixed mobility scenarios and NS-3 simulates wireless packet delivery over IEEE 802.11p/DSRC and 5G NR-V2X channels, including packet transmission rates, delays, jitters, dynamic routing information, and mobility-induced link changes.

To ensure complete reproducibility, we explicitly define NOVA's fundamental configurations before the presentation of results. For each target node, we create contextual candidates through random walks (with a walk length of $L = 4$, $M = 10$ walks per node, and $K_{cand} = 20$ candidate subgraphs). We then keep the top $k = 5$ contexts based on the intimacy score calculated using PageRank-style damping with $\alpha = 0.15$. For vehicular traces (VeReMi/CARLA), we use a rolling time window of $\Delta = 1$ s, and for flow datasets (CICIDS2017/UNSW-NB15/NSL-KDD), we use a rolling time window of $\Delta = 5$ s. Target-node anonymization is implemented by replacing the target node's feature vector with an all-zero vector, i.e., $x_i \leftarrow 0 \in \mathbb{R}^F$, within each sampled context subgraph, no mean vector or learnable [MASK] token is used. This makes reconstruction and structural validation depend on the neighborhood. The generative branch uses a one-layer GCN encoder and a one-layer GCN decoder with ReLU activation. The contrastive branch uses mean pooling to get the context embedding and a sigmoid bilinear discriminator to score node-context pairs. Negative pairs are made by changing the edge/feature of the contextual view. Hybrid scoring uses balanced fusion with $\lambda = 0.5$ and $\gamma = 0.5$. The decision threshold is set to $\tau = \tau^*$ selected on the validation split by maximizing $F1@ \tau$ (ties broken by lower FAR). To cut down on the random selection of contexts, final anomaly scores are averaged over $R = 5$ repeated sampling trials during inference.

B. Dataset

We use five diverse data sources (VeReMi, CARLA-based IoV traces, CICIDS2017, UNSW-NB15, and NSL-KDD) to assess the NOVA under a variety of IoV and network-security scenarios. VeReMi and CARLA include vehicle/V2X-focused traces, containing malicious activity, mobility, and communication events, whereas CICIDS2017 and UNSW-NB15 provide a variety of contemporary IP-based attack scenarios to test the cross-domain intrusion-detection capacity. NSL-KDD is included as a baseline to compare with previous IDS works. To facilitate fair comparisons, we map each of these diverse data sources into a common set of dynamic attributed graphs. At each time step t , we build $G_t = (V_t, E_t, X_t)$, where nodes represent IoV communication parties (vehicles, RSUs, hosts, or infrastructure services), edges model observed interactions, flows, or connectivity events, and node attributes quantify a range of behavioral features such as packet/byte rates, flow/messages count, inter-arrival statistics, protocol or service type distributions, delay/loss proxies, and mobility features when available. In VeReMi and CARLA, we map graph structures to V2V, V2I and V2N interactions; in CICIDS2017, UNSW-NB15, and NSL-KDD, we treat endpoints as if they were IoV communication endpoints and aggregate endpoint-level flow statistics.

To make it possible to reproduce the results, we turn each dataset into a series of windowed, dynamically attributed graphs $G_t = (V_t, E_t, X_t)$, using fixed-length time windows of length Δ (Table II shows the dataset-specific node/edge definitions, window sizes, and feature groups). Nodes represent entities that communicate (vehicles/RSUs for vehicular traces; endpoints/hosts for flow datasets), edges show "active links" that were seen in window t , and node attributes X_t are calculated as per-node aggregates of all events/flows that were assigned to that window via $t = \lfloor (T - T_0)/\Delta \rfloor$. If u and v interact at least once in the window, an edge $(u, v) \in E_t$ is created. For flow datasets, bidirectional flows are combined into a single undirected interaction, with optional edge aggregates like flow count and byte volume. In VeReMi and CARLA, active links come from V2X message exchanges and/or connectivity events (V2V/V2I/V2N). When both logs are available, message exchange is given priority, and connectivity is used to keep short-lived but physically plausible neighbors. We use a consistent feature template across datasets, using available subsets of: traffic intensity (packet/byte rates, message/flow counts), timing (inter-arrival mean/variance, burstiness proxies), protocol/service composition (e.g., TCP/UDP/ICMP fractions or service types), reliability proxies (loss/failed-connection ratios or error indicators), routing/control dynamics (vehicular traces), and mobility descriptors (speed/acceleration/heading-change for CARLA/VeReMi); all numeric features are normalized per dataset (z-score). We set $\Delta = 1$ s for vehicular traces and $\Delta = 5$ s for flow datasets unless otherwise noted. To explicitly synchronize it, each timestamp T record is assigned a window $t = \lfloor (T - T_0)/\Delta \rfloor$, and all the packets, flows and mobility/beacon updates which are mapped to the same t are considered to be concurrently observed to form $G_t = (V_t, E_t, X_t)$. We set Δ in a way that gathers several vehicular beacon chances and maintains local

mobility consistency, namely, we would take $\Delta = 1 \text{ s} = 10b$ in the case of CAM/BSM-style signaling with nominal period $b \approx 100 \text{ ms}$ (10 Hz), and $\Delta = 5 \text{ s}$ in the case of flow data. When a node is sampled in between two beacon intervals, and no new beacon-derived attribute is seen in the current window, we use a last-observation-carried-forward rule in the same horizon Δ ; formally, $x_i^{(m)}(t) = x_i^{(m)}(t^-)$ in the most recent valid observation t^- within that horizon, and otherwise the attribute is replaced by a neutral default. The sampling is therefore based on the representation on a synchronized window level as opposed to a single beacon at the instantaneous level, and therefore, the fact that beacon updates will be missed occasionally does not refute context selection. We also only synchronize asynchronous records by assigning timestamps to windows and treating interactions within a window as concurrent. Graphs are made for each dataset separately (no merging across datasets) and split into three-time windows: 70% for training, 10% for validation, and 20% for testing. This is done to avoid temporal leakage.

TABLE II
DATASET-TO-GRAPH MAPPING

Dataset	Nodes	Edges	Δ	Attributes
VeReMi	Vehicles/RSU s	V2X message exchange	1 s	Intensity, timing, routing
CARLA	Vehicles/RSU s	V2X exchange/connectivity	1 s	Intensity, timing, mobility
CICIDS2017	Endpoints/hos ts	Flows per pair (bidirectional consolidated)	5 s	Intensity, timing, protocol/service
UNSW-NB15	Endpoints/hos ts	Flows per pair (bidirectional consolidated)	5 s	Intensity, timing, protocol/service
NSL-KDD	Logical endpoints	Connections per pair	5 s	Intensity, timing, protocol/service

C. PERFORMANCE METRICS AND EVALUATION

We have chosen representative baselines that represent the key paradigms of IoV/CAV anomaly detection: CNVA [20], MKFA [22], IVNS [23], VADG [29], DST-IDS [35], and GATransformer [36]. They collectively include CNN-BiLSTM/VAE, attention-transformer, self-supervised transformer, GAN-based, and graph-based IDS families, and are compared using an identical data pipeline and metrics.

• Performance Comparison

Fig. 4 is a full comparison of all the baselines. Fig. 4(a) also demonstrates that NOVA performs best on all datasets, with a 98.7% accuracy, 97.9% precision, 98.3% recall, and 98.1% F1-score. The nearest competitor of the graph-based baselines is GATransformer, then DST-IDS, which validates the advantage of explicit spatio-temporal graph modeling. Nevertheless, NOVA continues to perform better due to the combination of intimacy-based context selection and GCN-based reconstruction with node-context contrastive validation. Fig. 4(b) further shows that NOVA maintains the highest F1-score across all datasets, ranging from 97.8% to 98.4%, while GATransformer ranges from 97.2% to 97.8%, DST-IDS from 96.9% to 97.4%, VADG from 95.9% to 96.5%, IVNS from 94.6% to 95.1%, MKFA from 92.6% to 93.3%, and CNVA from 91.0% to 91.5%. The respective FAR trends are reported in Fig. 4(c), with the lowest false-alarm rate of NOVA

between 3.8% and 4.5%. Comparatively, the graph-based baselines would be upper, 5.0%-5.6% of GATransformer and 5.8%-6.5% of DST-IDS, whereas the non-graph baselines have higher FAR values.

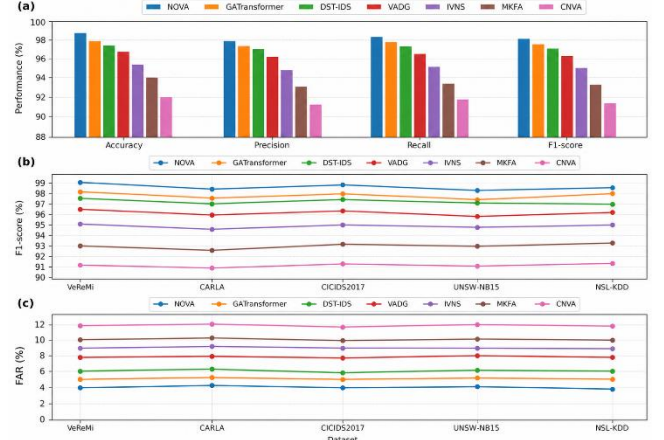


Fig. 4. Performance comparison of methods across the average results of all datasets.

TABLE III

PER-DATASET PERFORMANCE COMPARISON (%)

Dataset	Method	Accuracy	Precision	Recall	F1-Score
VeReMi	NOVA	98.9	98.2	98.6	98.4
VeReMi	GATransformer	98.1	97.4	98.2	97.8
VeReMi	DST-IDS	97.7	97.1	97.7	97.4
VeReMi	VADG	96.7	96.0	96.4	96.2
VeReMi	IVNS	95.4	94.6	95.1	94.9
VeReMi	MKFA	93.5	92.7	93.1	92.9
VeReMi	CNVA	91.8	91.0	91.5	91.2
CARLA	NOVA	98.6	97.8	98.1	97.9
CARLA	GATransformer	97.5	96.9	97.7	97.3
CARLA	DST-IDS	97.1	96.6	97.2	96.9
CARLA	VADG	96.4	95.8	96.0	95.9
CARLA	IVNS	95.2	94.5	94.8	94.6
CARLA	MKFA	93.2	92.4	92.8	92.6
CARLA	CNVA	91.5	90.9	91.1	91.0
CICIDS2017	NOVA	98.8	98.0	98.4	98.2
CICIDS2017	GATransformer	98.0	97.3	98.1	97.7
CICIDS2017	DST-IDS	97.6	97.0	97.7	97.3
CICIDS2017	VADG	97.0	96.3	96.7	96.5
CICIDS2017	IVNS	95.7	95.0	95.3	95.1
CICIDS2017	MKFA	93.9	93.0	93.6	93.3
CICIDS2017	CNVA	92.0	91.3	91.7	91.5
UNSW-NB15	NOVA	98.5	97.6	98.0	97.8
UNSW-NB15	GATransformer	97.4	96.8	97.6	97.2
UNSW-NB15	DST-IDS	97.0	96.5	97.3	96.9
UNSW-NB15	VADG	96.6	95.9	96.2	96.0
UNSW-NB15	IVNS	95.3	94.7	95.0	94.8
UNSW-NB15	MKFA	93.6	92.8	93.2	93.0
UNSW-NB15	CNVA	91.7	91.1	91.4	91.2
NSL-KDD	NOVA	98.7	97.9	98.3	98.1
NSL-KDD	GATransformer	97.8	97.1	97.9	97.5
NSL-KDD	DST-IDS	97.4	96.8	97.5	97.1
NSL-KDD	VADG	96.9	96.2	96.5	96.3
NSL-KDD	IVNS	95.6	94.9	95.2	95.0
NSL-KDD	MKFA	93.8	92.9	93.4	93.2
NSL-KDD	CNVA	91.9	91.2	91.6	91.4

Table III shows how well VeReMi, CARLA, CICIDS2017, UNSW-NB15, and NSL-KDD did on each dataset (Accuracy, Precision, Recall, and F1-Score). Its accuracy is around 98.5–98.9%, and its F1-Score is around 97.8–98.4%, which shows that it can detect things well and consistently across different situations.

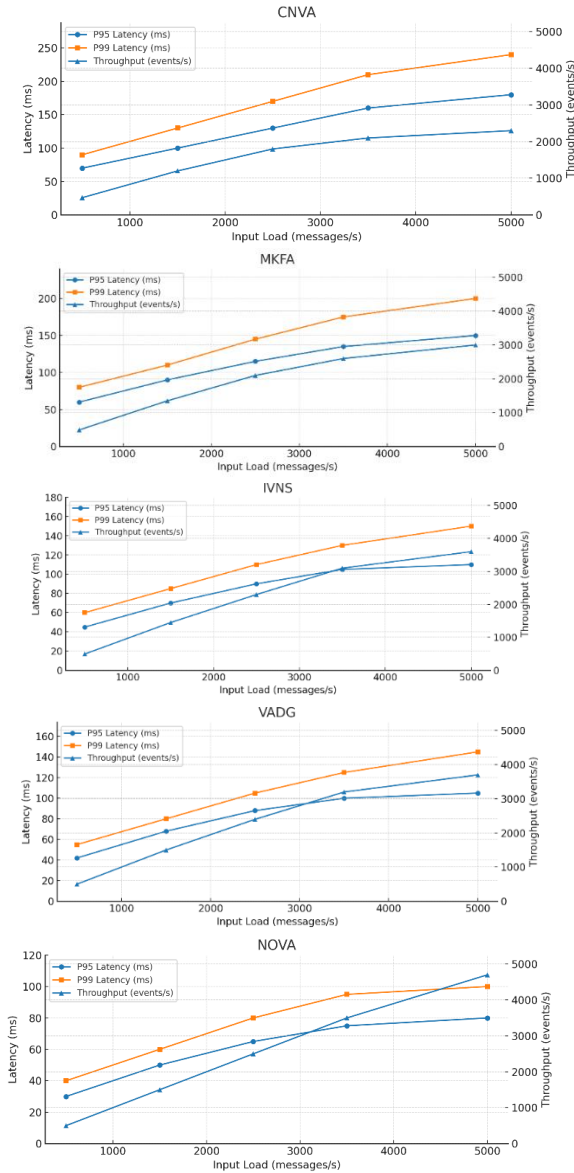


Fig. 5. QoS under variable input load (500→5k messages/s): per-method P95/P99 end-to-end latency (left Y) and throughput (right Y).

Since this may be biased by the corpus, the latency and throughput data reported in Fig. 5 are averages over all datasets. P95 latency is used to encode high load latency behavior and P99 latency is used to capture tail latency behavior, which is critical for such applications as emergency messages in IoV applications, braking decisions, and multi-hop relay V2X. As the input load is increased to 5 k msg/s, the delay growth of CNVA is the highest among the three, with P95 growing from ~70 to 180 ms, and P99 growing from ~90 to 240 ms, and the events/s is still ~2.3 k events/s. The overhead of feature-engineering and distillation, however, causes high tail latency in MKFA, whereas both throughput and tail latency are low. IVNS and VADG have lower latency and higher throughput, with IVNS reaching P95/P99 of ~110/~150ms and ~3.6k events/s, and VADG reaching ~105/~145ms and ~3.7k events/s at a high load. NOVA's latency profile is quite flat, with P95 rising from ~30 to 80 ms,

and P99 from ~40 to 100 ms, and a sustained throughput of ~4.7 k events/s at 5 k msg/s.

The end-to-end latency for real-time detection includes three stages: (i) graph orchestration, which includes graph feature extraction, time-windowing, and incremental graph updates; (ii) context selection includes candidate generation, intimacy scoring, and top-k selection; and (iii) model inference includes GCN encoder-decoder passes, contrastive discrimination, and hybrid scoring. The subgraph-local inference cost is $O(Rk | V_s | FH)$, and the final scoring costs $O(R(F + H))$. To minimize overhead during mobility churn, RSUs incrementally update local graph neighborhoods and are then able to reuse cached neighbor statistics for intimacy scoring without reconstructing the entire graph.

• False Alarms under Mobility Churn

We examine the urban mobility churn in QoS evaluation: FAR and F1@ τ in Fig. 6. The best performance is achieved by NOVA, where the FAR is 0.05 and the F1@ τ is 0.98, indicating that it is very robust to speed changes, link dynamics and handoffs. VADG outperforms with FAR = 0.08 and F1@ τ = 0.95, and IVNS achieves F1 = 0.94 and FAR = 0.09, but has no structural adaptability. The handcrafted features cause MKFA to perform moderately (0.10, 0.93) and the lack of structural awareness causes CNVA to perform worse (0.12, 0.91).

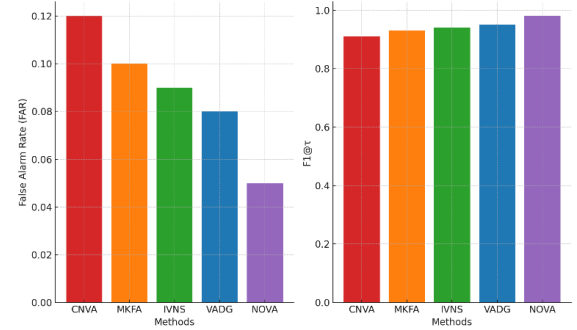


Fig. 6. FAR and F1-score at threshold τ (F1@ τ).

• Structural Attack Robustness

Fig. 7, it shows that Sybil identities, bogus neighbor insertions, and link perturbations are effective structural attacks on IoV communication graphs. NOVA has the best AUROC of 0.99, AUPRC of 0.98 and the lowest misclassification error of 0.05. VADG and IVNS are also good, but do not show an explicit use of structural reasoning and focus primarily on sequential patterns.

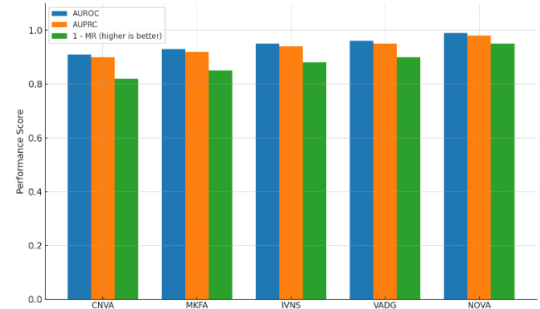


Fig. 7. Structural attack robustness across methods, showing AUROC, AUPRC, and inverse MR (1-MR).

- **Label Efficiency / Label-Free Operation**

The label efficiency is measured on CICIDS2017 and UNSW-NB15, where 10% of the data is labeled for reliable labels for controlled ablation, as shown in Fig. 8. The AUROC of CNVA and MKFA drops significantly as the number of labels decreases, and eventually becomes below AUROC 0.70 when there are no labels. Although limited supervision, VADG stays stable with little improvement, IVNS, thanks to self-supervised transformer learning, remains above 0.90. Without label information, NOVA gets the best fit at 0.89, and keeps increasing steadily to 0.97 at 10% labels.

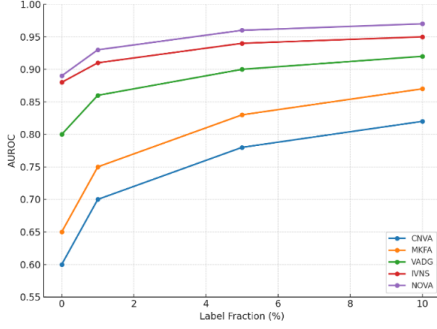


Fig. 8. AUROC vs. label fraction across methods under CICIDS2017 and UNSW-NB15 datasets with labels gradually ablated.

- **Concept Drift & Online Adaptation**

These datasets were then used to simulate progressive drift with various conditions of vehicles and clouds. Based on Fig. 9, NOVA is the most “stable”, as it has suffered only a ~4% drop in F1, and IVNS a ~6% drop in F1. VADG experiences a drop of ~14% when the traffic is not stationary, while MKFA experiences a drop of 20% because of handcrafted features. The performance is worst in CNVA with a drop of more than 24% due to the fact that the feature is static and the embeddings are static, which is bound to a sensor.

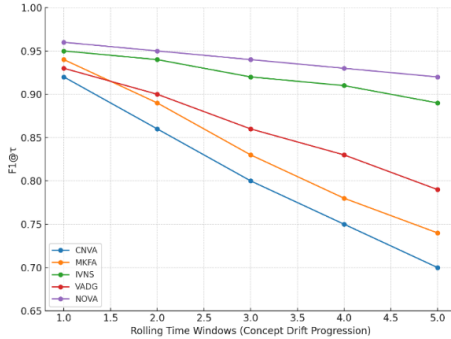


Fig. 9. Concept drift and online adaptation.

- **Scalability with Fleet/Graph Size**

There are differences in scalability in large vehicle networks, as illustrated in Fig. 10. NOVA does best on 50,000 vehicles, processing up to 25,000 events/s, and achieving a latency of less than 40 ms and memory of less than 2 GB. MKFA and CNVA scale poorly, achieving 2,800 and just over 2,000 events/s, respectively, while CNVA consumes over 120 ms latency and 4 GB of memory.

- **Adversarial & Poisoning Robustness**

Fig. 11 compares the robustness with respect to FGSM, PGD and poisoning attacks. Under FGSM (PGD), NOVA has the

highest level of resistance, with an AUROC of ≈ 0.95 (≈ 0.93); and the backdoor success rate is around 10%. IVNS and VADG are still moderately powerful (≈ 0.85 - 0.88 AUROC), but IVNS is still prone to sequence-level triggers and VADG to poisoning instability. The performance of the two algorithms that do not use adversarial regularization, MKFA and CNVA, is weakest with AUROCs less than ≈ 0.85 . Overall, NOVA has an AUROC of about 10% higher and 2–3 times lower poisoning success, making it more appropriate for adversarial IoV deployments.

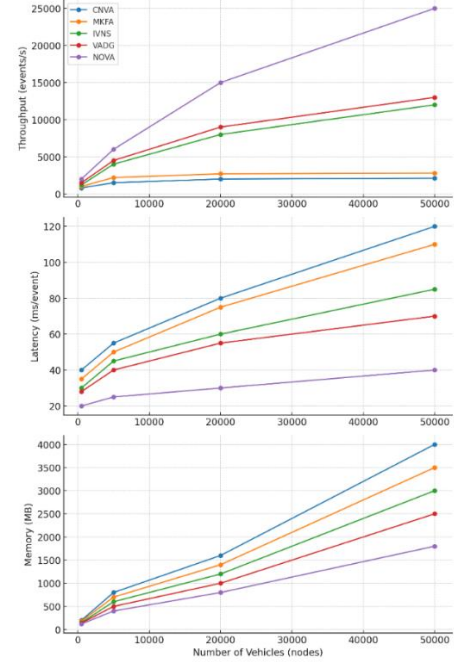


Fig. 10. Throughput, latency, and memory usage for CNVA, MKFA, IVNS, VADG, and NOVA as the number of vehicles increases from 500 to 50,000 in NS-3 with RSU aggregation.

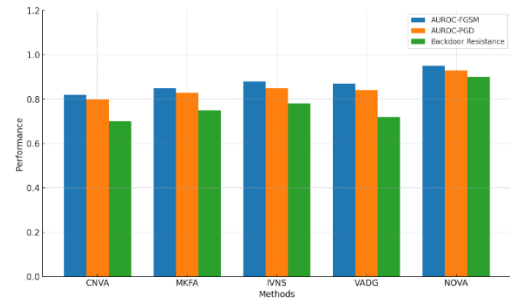


Fig. 11. Adversarial and poisoning robustness across methods.

- **Cross-Modality Generalization**

The differences in modality transfer are obvious as seen in Fig. 12. The smallest generalization gaps (~ 0.05 sensor-to-comm and ~ 0.04 comm-to-sensor) are obtained by NOVA, which can simultaneously learn a combination of attribute vectors and structural embeddings in its graph abstraction. The lower gaps follow after IVNS ($\sim 0.12/0.11$), but the stability is affected by the topology changes. However, the gaps for CNVA ($\sim 0.20/0.18$) are the largest, attributable to the reliance on engineered features, and the reliance on a LiDAR-like sensor setting means that the accuracy is not maintained when transferring beyond that setting.

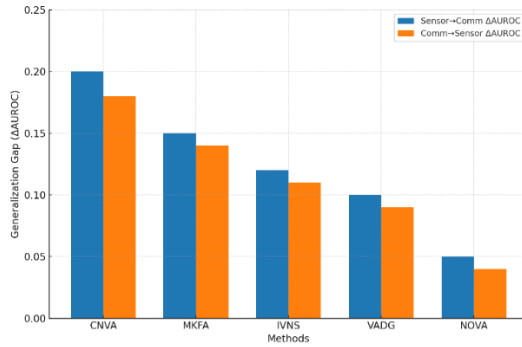


Fig. 12. Cross-modality generalization gap ($\Delta AUROC$) when transferring from sensor-heavy datasets (LiDAR/CARLA) to communication-heavy datasets.

• Absolute Context-Selection Latency Analysis

The absolute wall-clock latency of the isolated context-selection stage in the same settings was reported in Fig. 13 and the absolute latency of the RandomWalk baseline was compared to the intimacy-guided selection of NOVA. RandomWalk baseline can only generate the candidates through random-walking, but NOVA can generate the candidates and then score them using intimacy and select the top-k. The extra overhead of the intimacy-guided selection is not very large in absolute terms: the mean latency becomes 5.36 ms and the P95 and P99 latencies become 7.41 ms and 8.94 ms, respectively. These correspond to absolute increases of 1.08 ms, 1.36 ms, and 1.63 ms, and relative overheads of 25.2%, 22.5%, and 22.3%, respectively.

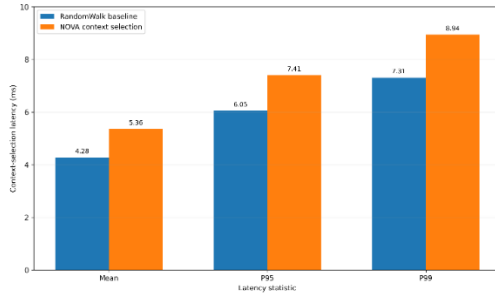


Fig. 13. Absolute context-selection latency comparison between the RandomWalk baseline and NOVA's intimacy-guided selection under identical settings.

• Adaptive Joint-Objective Multi-Context Attack Robustness

Fig. 14 provides the Attack Success Rate (ASR) of four increasingly robust adaptive adversaries with perturbation budgets of 2 to 10%. The weakest of the two attackers is the reconstruction-only attacker, whose ASR grows slowly, 6 to 22 percent, which means that it is not enough to suppress the generative signal to be able to evade NOVA with a high probability. The alignment-only attacker is more potent, and ASR rises by 8 to 29, and it is clear that the attack on the node-context similarity is more effective than the attack on the reconstruction error, yet it is not strong enough to outsmart the hybrid detector. The joint multi-context attacker is significantly stronger when both objectives are being optimized: the feature-only joint multi-context attacker achieves ASR 12% at 2% budget and 43% at the 10% budget, a 21% and 14% improvement over the reconstruction-only

attacker and the alignment-only attacker, respectively. The best is the joint multi-context attacker, where there is simultaneous perturbation of features and topology, which raises its ASR between 15 and 51.

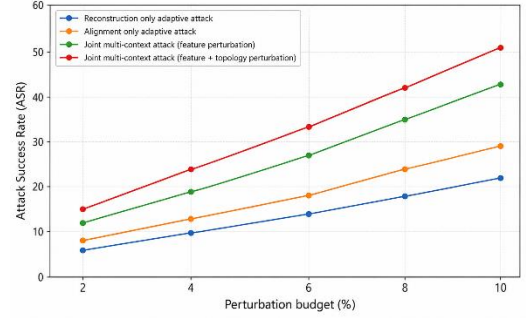


Fig. 14. ASR under increasingly strong adaptive attacks.

• Sparse-Context Fallback and Performance

Fig. 15 demonstrates how decreased availability of context affects NOVA with a sparse-context fallback policy, where the model applies all available high-intimacy contexts and averages the score by this actual number. Starting from the default regime of $m = 5$, NOVA achieves $F1@τ = 0.980$, $AUROC = 0.990$, and $FAR = 0.040$. The degradation is small when one of the contexts is not present ($m = 4$), with $F1@τ = 0.976$, $AUROC = 0.987$, and $FAR = 0.043$, or a loss of just 0.4 percentage points in $F1$ and 0.3 points in $AUROC$. At $m = 3$, the performance is good with $F1@τ = 0.969$, $AUROC = 0.981$, and FAR is slightly greater, at 0.048. More noticeable degradation appears only under severe sparsity: at $m = 2$, $F1@τ$ drops to 0.955, $AUROC$ to 0.970, and FAR rises to 0.058; at $m = 1$, the model reaches $F1@τ = 0.934$, $AUROC = 0.951$, and $FAR = 0.071$.

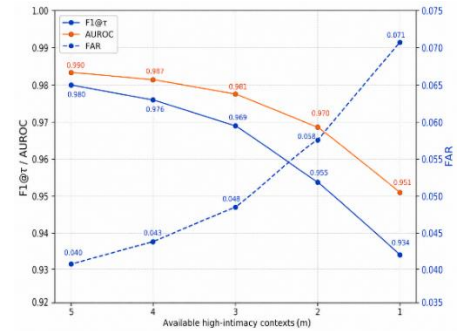


Fig. 15. Performance degradation under reduced context availability using NOVA's sparse-context fallback. When fewer than the default $k = 5$ high-intimacy contexts are available, the model uses all available contexts and normalizes by the actual count.

As can be seen in Fig. 16, NOVA is the strongest when stressed. NOVA achieves the lowest $F1@τ$ (0.982→0.965) and the lowest FAR increase (0.038→0.055) compared to DST-IDS and GATransformer, which shows that it has the least mobility churn. NOVA also shows improved $AUROC$ (0.990→0.968) and reduced MR (0.045→0.078) under a structural perturbation. In concept drift, both baselines (0.981→0.956 and 0.975) degrade less and recover from the update faster than NOVA (0.981→0.956 and 0.975) does.

As can be seen in the results shown in Fig. 17, NOVA is the most efficient in terms of deployment when realistic IoV workloads are taken into account. NOVA has the lowest P99

latency as load progressively increases from 500 to 5000 messages/s, increasing from ~ 38 to ~ 95 ms, which is better than DST-IDS (~ 42 – 155 ms) and GATransformer (~ 44 – 165 ms). NOVA also achieves the highest throughput across 500–50,000 nodes ($\sim 8,500$ – $25,500$ events/s) while using the least memory at 50,000 nodes (~ 1.85 GB vs. 2.80 GB and 3.10 GB).

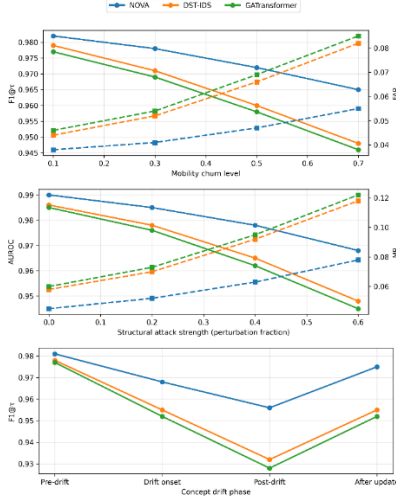


Fig. 16. Robustness stress test under realistic IoV conditions: (a) churn effects on $F1@τ/FAR$, (b) structural attacks on $AUROC/MR$, and (c) concept drift effects on $F1@τ$.

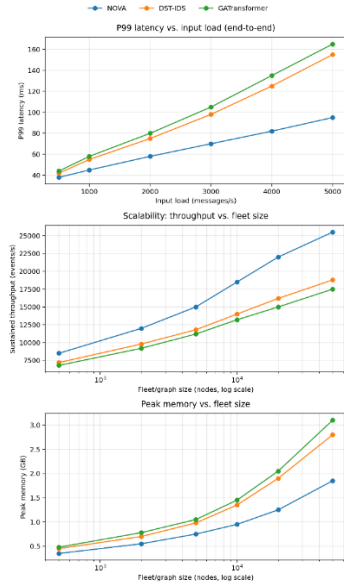


Fig. 17. Deployment efficiency of NOVA vs. GNN baselines: (a) P99 latency under increasing load, (b) throughput across fleet/graph sizes, and (c) peak memory usage across fleet/graph sizes.

D. Complexity Analysis

Let $N = |V|$ be the number of nodes in the global IoV graph, K_{cand} the number of candidate subgraphs per target node, L the random-walk length, k the number of selected contexts, R the number of repeated inference trials, and $|V_s|$, $|E_s|$, F , and H the average subgraph nodes, subgraph edges, input feature dimension, and hidden dimension, respectively. During inference, candidate generation costs $O(K_{\text{cand}}L)$, intimacy

scoring costs $O(K_{\text{cand}}(|V_s| + |E_s|))$, and top- k selection costs $O(K_{\text{cand}} \log k)$. For each selected context, the one-layer GCN encoder-decoder costs $O(|V_s|FH + |E_s|(F + H))$, while pooling, bilinear node-context discrimination, reconstruction-distance computation, and score fusion add $O(|V_s|H + H^2 + F)$. Therefore, the per-node inference complexity is $O(R[K_{\text{cand}}(L + |V_s| + |E_s| + \log k) + k(|V_s|FH + |E_s|(F + H) + |V_s|H + H^2 + F)])$. Under bounded RSU-local contexts, K_{cand} , L , k , R , $|V_s|$, and $|E_s|$ are fixed deployment constants. Hence, the per-node inference cost is independent of the global fleet size N , and fleet-level inference scales linearly as $O(N)$. This formal inference-time complexity supports NOVA's real-time execution on the RSU-side and vehicular-edge infrastructure.

E. Discussion

NOVA can be deployed in an edge-cloud hierarchy to achieve a balance of latency, bandwidth and computational cost for real IoV systems. On the edge, RSUs are capable of windowed feature aggregation, intimacy-driven subgraph sampling and lightweight inference based on a GCN and hybrid anomaly scoring, allowing to detect anomalies in real-time without sharing raw telemetry. Cloud provides long-term statistics, calibrates thresholds, provides cross-RSU consistency, retrains the model, tunes hyperparameters, and distills global models for heterogeneous RSUs. RSUs should be located at high-observability hot-spots (intersections, highway merges, urban corridors), and overlap should be provided to enhance handoff and cross-RSU scoring. Communication overheads are minimized by having RSUs send only a summary and a few graph updates, and the cost of the uplink varies roughly as $O(|V_{\text{RSU}}| \cdot D) + O(|E_{\text{RSU}}|)$ per window, after sparsification, quantization and batching. Updates of the model are made using a hybrid approach: frequent lightweight threshold adjustments, and infrequent changes of the model when retraining is scheduled or when a drift indicator is activated. To guarantee the same detection, in bandwidth-limited settings, model deltas or distilled student models can be downloaded to the RSUs.

Also, our experiments focus on standard evasion and poisoning; however, it is crucial to account for adaptive adversaries that exploit both attributes and topology to concurrently diminish reconstruction error and enhance node-context alignment. Transferable adversarial attacks add another form of black-box threat where the attacker trains a surrogate IDS on a similar set of data of vehicles and applies the resulting perturbations to NOVA. NOVA can be susceptible to these attacks in part due to its GCN encoder that projects node attributes and local topology into latent embeddings, and trained surrogate graph models that learn partially correlated feature sensitivities on similar V2X distributions. But in NOVA, we also have mechanisms that are capable of lowering direct transferability. Target-node anonymization makes sure that the detector does not just use the raw feature vector of the target node, but rather requires reconstruction and validation to be based upon neighborhood context, which has the ability to smooth sensitivities of shortcut-based features. Furthermore, intimacy-based multi-context testing involves transferred perturbations to be

consistent among a set of highly relevant subgraphs, whereas hybrid generative-contrastive scoring involves the attacker minimizing reconstruction error and node-context compatibility at the same time. Thus, despite the fact that NOVA is not resistant to transferable adversarial attacks, its context-conditioned and dual-path representation does not ensure the direct perturbation transfer in comparison with the IDS models that are based solely on features.

NOVA is hard to evade as a node is evaluated in multiple intimacy-selected contexts using both GCN-based profile synthesis and node-node contrast in an intimacy-driven context. So, adversaries must preserve their behavior across multiple relevant neighborhoods as topologies shift due to mobility. Although poisoning or collusion attacks (e.g., Sybil) can still impact the learned normal profile, NOVA mitigates shortcut learning by anonymizing target nodes and assessing valid (as opposed to malicious) contexts. In practice, robustness can be further enhanced using a sliding-window retraining, trusted-trace calibration, cross-RSU correlation, reputation, and plausibility checks in the physical layer.

VI. CONCLUSION AND FUTURE WORK

In this paper, we introduced a self-supervised graph-based framework for IoV anomaly detection, called NOVA. NOVA is designed to detect behavioral and topological anomalies by engaging in both graph structure validation via contrastive learning and attribute reconstruction using the GCN framework while simultaneously sampling graphs from the graph domain with guidance from the intimacy. NOVA achieves high detection accuracy, F1 score, AUROC/AUPRC, low FAR and latency, scalable and high adversarial robustness and outperforms baselines in all these metrics across extensive evaluations performed on VeReMi, CARLA-IoV, CICIDS2017, UNSW-NB15, and NSL-KDD. NOVA is effective in the presence of mobility churn, structural attacks, concept drift, label scarcity and poisoning/evasion attempts, and is scalable linearly and maintains near real-time tail latency. The results show that the self-supervised graph learning with anonymized contextual reconstruction and hybrid scoring is a viable and reliable learning paradigm for ensuring safety in safety-critical IoV/V2X environments.

We tested NOVA in real time using a SUMO-NS3 co-simulation IoV testbed with RSU/cloud hardware. The results were promising, but there are still some problems. First, the testbed uses simulated mobility and network stacks, so more field testing with V2X traffic over the air is needed to make sure it works in the real world. Second, even though we add examples of behavioral and structural anomalies (like Sybil, DoS, and link perturbations), the adversarial landscape is always changing. This calls for more stress testing and constant model updates to protect against multi-stage, stealthy, and adaptive attacks. Future directions will also test black-box transferable adversarial attacks on independently trained surrogate graph models and explore randomized latent feature selection or feature-subspace ensembling as other defense strategies. Thirdly, NOVA uses subgraph-local reconstruction and node-context validation, which could turn out to be less reliable in scenarios characterized by sparse connectivity or high mobility, where the neighborhood structure is unstable

and thresholds might be more difficult to calibrate. Future work will involve: (i) Federated and edge distributed NOVA for privacy preserving cross fleet learning; (ii) Temporal graph extensions for dynamic V2X interactions and short-lived anomalies; (iii) Cross-modality fusion with LiDAR, GPS, and radar; (iv) Trust/reputation integration for risk-aware mitigation; (v) Adaptive thresholding and online calibration against poisoning and evasion; and (vi) hardware efficient acceleration for real-time RSU and in-vehicle deployment.

SUPPLEMENTARY MATERIAL

Additional ablation studies, sampling and anonymization analyses, parameter-sensitivity evaluations, robustness results, and real-time deployment details are provided in the Supplementary Material.

DECLARATION OF COMPETING INTEREST: NA

REFERENCES

- [1] I. U. Din, K. H. Khan, A. Almogren, and M. Guizani, "Machine Learning for Trust in Internet of Vehicles and Privacy in Distributed Edge Networks," *IEEE Internet of Things Journal*, 2025.
- [2] Y. Fang *et al.*, "Anomaly diagnosis of connected autonomous vehicles: A survey," *Information fusion*, vol. 105, p. 102223, 2024.
- [3] H. I. Ali, H. Kurunathan, M. H. Eldefrawy, F. Gruian, and M. Jonsson, "Navigating the challenges and opportunities of securing internet of autonomous vehicles with lightweight authentication," *IEEE Access*, 2025.
- [4] R. Wang *et al.*, "A Glimpse of Physical Layer Security in Internet of Vehicles: Joint Design of the Transmission Power and Sensing Power," *IEEE Transactions on Vehicular Technology*, 2025.
- [5] T. Alshammari and I. Mahgoub, "Nature-inspired algorithms in the internet of vehicles: A survey and analysis," *IEEE Internet of Things Journal*, 2025.
- [6] J. R. V. Solaas, E. Mariconti, and N. Tuptuk, "Systematic literature review: Anomaly detection in connected and autonomous vehicles," *IEEE Transactions on Intelligent Transportation Systems*, 2024.
- [7] H. Huang, P. Wang, J. Pei, J. Wang, S. Alexanian, and D. Niyato, "Deep learning advancements in anomaly detection: A comprehensive survey," *IEEE Internet of Things Journal*, 2025.
- [8] A. Sakhnevych, N. Pasquino, and G. Sperli, "Design of a machine learning approach to anomaly detection in tyre-road interaction," *IEEE Access*, 2025.
- [9] L. Zheng, T. Feng, Z. Xie, X. Li, and C. Su, "BARM: Blockchain-Assisted Anonymous Authentication and Reputation Management for Mobile Crowdsensing in Internet of Vehicles," *IEEE Internet of Things Journal*, 2025.
- [10] P. K. R. Lebak, L. Gao, Y. Zhang, Z. Li, Y. Liu, and T. Arafat, "Cybersecurity-focused anomaly detection in connected autonomous vehicles using machine learning," in *International Conference on Transportation and Development 2025*, 2025, pp. 566-580.
- [11] Y. Zhang *et al.*, "A cooperative vehicle-road system for anomaly detection on vehicle tracks with augmented intelligence of things," *IEEE Internet of Things Journal*, vol. 11, no. 22, pp. 35975-35988, 2024.
- [12] S. Mishra, N. Sengar, and D. Har, "A Secure, Blockchain-Enabled Vehicular Sensor Communication Protocol With Deep Learning-Assisted Anomaly Detection," *IEEE Intelligent Transportation Systems Magazine*, 2025.
- [13] X. Wang *et al.*, "Hyperspectral Anomaly Detection Using Dual-Branch Network Based on Frequency Domain Learning," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 2025.

- [14] S. Baccari, M. Haddad, H. Ghazzai, H. Touati, and M. Elhadeif, "Anomaly detection in connected and autonomous vehicles: A survey, analysis, and research challenges," *IEEE access*, vol. 12, pp. 19250-19276, 2024.
- [15] D. Wang, L. Ren, X. Sun, L. Gao, and J. Chanussot, "Non-local and local feature-coupled self-supervised network for hyperspectral anomaly detection," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 2025.
- [16] Q. Cheng, K. Hong, K. Huang, and Z. Liu, "Evaluating Effectiveness and Identifying Appropriate Methods for Anomaly Detection in Intelligent Transportation Systems," *IEEE Transactions on Intelligent Transportation Systems*, 2025.
- [17] Z. Song, Y. Tao, Z. Hua, S. Wang, G. Pan, and J. An, "Generative Artificial Intelligence-Empowered Multidomain Internet of Vehicles Systems: Scalability, Efficiency, and Suitability," *IEEE Vehicular Technology Magazine*, 2025.
- [18] J. Boone, T. Seyfi, and F. Afghah, "A Joint Reconstruction-Triplet Loss Autoencoder Approach Towards Unseen Attack Detection in IoV Networks," *IEEE Internet of Things Journal*, 2025.
- [19] E. Nowroozi, M. Mohammadi, A. Rahdari, R. Taheri, and M. Conti, "A random deep feature selection approach to mitigate transferable adversarial attacks," *IEEE Transactions on Network and Service Management*, 2025.
- [20] N. Sboui, H. Ghazzai, M. Haddad, M. Elhadeif, and G. Setti, "Unsupervised Hybrid VAE-Based Anomaly Detection for Vehicle Onboard LiDAR Sensors," *IEEE Access*, 2025.
- [21] P. K. Tiwari *et al.*, "A secure and robust machine learning model for intrusion detection in internet of vehicles," *IEEE Access*, 2025.
- [22] P. Cheng, S. Liu, Z. Wu, L. Tan, and G. Liu, "MKF-ADS: Multi-Knowledge Fusion Based Anomaly Detection System in Vehicular Control Area Networks," *IEEE Transactions on Vehicular Technology*, 2025.
- [23] J. Cao *et al.*, "Anomaly detection for in-vehicle network using self-supervised learning with vehicle-cloud collaboration update," *IEEE Transactions on Intelligent Transportation Systems*, vol. 25, no. 7, pp. 7454-7466, 2024.
- [24] L. Zhao *et al.*, "Generative abnormal data detection for enhancing cellular vehicle-to-everything-based road safety," *IEEE Transactions on Green Communications and Networking*, vol. 8, no. 4, pp. 1466-1478, 2024.
- [25] A. Singh and H. Rathore, "Advancing connected vehicle security through real-time sensor anomaly detection and recovery," *Vehicular Communications*, vol. 52, p. 100876, 2025.
- [26] X. Li, C. Xie, Z. Zhao, C. Wang, and H. Yu, "Anomaly detection algorithm of industrial internet of things data platform based on deep learning," *IEEE Transactions on Green Communications and Networking*, vol. 8, no. 3, pp. 1037-1048, 2024.
- [27] P. Mansourian, N. Zhang, A. Jaekel, and M. Kneppers, "Deep learning-based anomaly detection for connected autonomous vehicles using spatiotemporal information," *IEEE Transactions on Intelligent Transportation Systems*, vol. 24, no. 12, pp. 16006-16017, 2023.
- [28] F. Van Wyk, Y. Wang, A. Khojandi, and N. Masoud, "Real-time sensor anomaly detection and identification in automated vehicles," *IEEE Transactions on Intelligent Transportation Systems*, vol. 21, no. 3, pp. 1264-1276, 2019.
- [29] S. Devika, R. R. Shrivastava, P. Narang, T. Alladi, and F. R. Yu, "Vadgan: An unsupervised gan framework for enhanced anomaly detection in connected and autonomous vehicles," *IEEE Transactions on Vehicular Technology*, vol. 73, no. 9, pp. 12458-12467, 2024.
- [30] Z. Yang *et al.*, "Anomaly detection against GPS spoofing attacks on connected and autonomous vehicles using learning from demonstration," *IEEE Transactions on Intelligent Transportation Systems*, vol. 24, no. 9, pp. 9462-9475, 2023.
- [31] Z. He, Y. Chen, D. Zhang, H. Zhang, and M. Liu, "MF-CANN: A Novel Anomaly Detection Method for Connected and Automated Vehicles," *IEEE Transactions on Industrial Cyber-Physical Systems*, vol. 2, pp. 588-596, 2024.
- [32] Z. He, Y. Chen, H. Zhang, and D. Zhang, "WKN-OC: A new deep learning method for anomaly detection in intelligent vehicles," *IEEE Transactions on Intelligent Vehicles*, vol. 8, no. 3, pp. 2162-2172, 2023.
- [33] F. Khanmohammadi and R. Azmi, "Time-series anomaly detection in automated vehicles using d-cnn-lstm autoencoder," *IEEE Transactions on Intelligent Transportation Systems*, vol. 25, no. 8, pp. 9296-9307, 2024.
- [34] Z. Zhang, Y. Yao, W. Hutabarat, M. Farnsworth, D. Tiwari, and A. Tiwari, "Time series anomaly detection in vehicle sensors using self-attention mechanisms," *IEEE Transactions on Intelligent Transportation Systems*, 2024.
- [35] G. A. Al-Absi, Y. Fang, A. A. Qaseem, and H. Al-Absi, "DST-IDS: Dynamic spatial-temporal graph-transformer network for in-vehicle network intrusion detection system," *Vehicular Communications*, vol. 55, p. 100962, 2025.
- [36] S. Wang, Y. Zhao, X. Fu, H. Si, W. Wang, and L. Xue, "A class-imbalance-aware intrusion detection system based on spatiotemporal graph neural networks for software-defined vehicles," *Journal of Information Security and Applications*, vol. 97, p. 104340, 2026.



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